

Xiaowei Li

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OBJECTIVE Seeking a software engineer position on the Schwab Market Data team

REASON: In the AI era, developers increasingly act as orchestrators directing AI agents. Deep hands-on experience with systems I built and maintained (market data applications) is a real advantage: it lets me quickly verify and correct AI-generated work.

SKILLS & HIGHLIGHTS

- 11 years C++, 7 years Python and C#/.NET, .NET Core, 1 year Java experience at Schwab
- Good problem solving, troubleshooting, bug fixing, and analytical skills
- GitHub/VSCode Copilot, MCP (Atlassian, GitHub), prompt engineering, agent skills
- Multithreading, object-oriented programming, RESTful API, IBM MQ, RabbitMQ, Certificate management
- Agile software practices, JIRA, Scrum, Sprint, Confluence, CI/CD Atlassian tools
- Git, GitHub, cvs, svn, Nuget, Bitbucket, Makefile, Msbuild
- WireShark, Postman, Bruno, Firebug, valgrind, gdb, Visual Studio, VS Code, vi, vim
- Shell script, bash, csh, ssh, socket, tcp/ip, udp, ssl, html, xml, css, json, snmp, mib
- SQL Server 2022, PostgreSQL, Aerospike, SQLite, MySQL, SSMS
- PCF, Linux/Ubuntu/Debian/CentOS, RHEL, Windows Server 2022, VirtualBox, Vmware
- UT-Austin Electrical & Computer Engineering background

EXPERIENCE

Senior/Staff Software Engineer, Charles Schwab, Austin, TX, 8/2015 ~ now

- **Trading OM team**
- Developed esm-data-loader, a Java-based program to load ESM data to PostgreSQL, including daily file loading and MQ intraday data loading; developed corm-listener, a Java-based Web API program to retrieve/save trading attributes from/to Aerospike.
- Trading software maintenance, OneOES, ASOS, ESD. Order entry/cancel/modify, order viewing, account viewing, Splunk log reduction, LaunchDarkly feature flag adoption.
- **Market Data team**
- Lead developer of interest-only OPRA CDA, reduced memory/CPU usage by 80%. Lead developer for interest-only OPRA CDA migration to AXIOM source.
- Lead developer of InfoQuotePublisher, a Cartman-based feed that combines BATS and delayed CTA/UTP data into a lower-cost product.
- Lead developer of spos-file-loader, which loads ESM data into SQL Server; maintained ESMMBusService for ESM intraday events loading to SQL.
- Lead developer of market-data-reference, daily symbol mapping update for cartman, MSS, etc.
- Bug fixing, troubleshooting in multiple market data systems, strong working relationships with the team.
- Market Data Applications, development and bug fixing, Linux, bash, putty, Scrum, Sprint, Agile.
- Intensive C++ C#/.Net Programming, STL, boost, OOP, API, Git, Bamboo, Bitbucket, JIRA, Confluence, Nexus, Nuget, CI/CD Atlassian tools.
- Open source message broker MQTT, C++11, RESTful API, tool development with python. Solaris, SolAdmin.
- Schwab News App (Reuters API and other vendors) maintenance and bug fixing, new feature development using C#, code migration.

Software Engineer, Contractor, Charles Schwab, Austin, TX, 12/2014 ~ 7/2015

- Market Data Applications, started with NewsSystem, then moved on to CDA and Feed Handler maintenance.

Software Engineer, Geist Global, ITWachDogs was sold to Geist, Austin, TX, 2/2011 ~ 12/2014

- Object-oriented, multithreaded, embedded development experience. Intensive C/C++ programming in developing new generation smart Power Distribution Unit (PDU) with ARM9 FreeScale MX1. Debian, TCP/IP stack, socket, IPv6, Net-SNMP v1 v2c v3, python, json, JavaScript. Used Wireshark to troubleshoot network issues. Bug fixing and product release based on Linux, gcc, g++, gdb.

- Implemented and debugged driver code for MACPHY, serial port, SPI.
- Ruby experience with web applications. Implemented and debugged GUI using firebug with Javascript and json. MediaWiki, asciidoc.
- Architect of company's continues integration tool. Designed and implemented fully automated build system and GUI testing tool. Using Jenkins and Selenium WebDriver 2.0 for automated GUI testing, wrote python test scripts for GUI functional testing. Test-driven and Agile experience.

Software Developer, ITWatchDogs Inc, Austin, TX, 4/2008 ~ 2/2011

- Code migration, bug fix, and legacy code maintenance of ITWatchDog WxGoose II series, firmware that runs a web server on ARM7 that collects multiple sensors (temperature, humidity, airflow, door position etc) data in Data Centers. Used debugging tools such as gdb and valgrind.
- Experience with real-time OS, C/C++, JTAG, Cygwin, serial bus, I2C, SPI, Dallas 1-wire bus, ARM, SNMP, MIB, TCP/IP, LWIP, UDP, WireShark, Microchip, Interrupt, Python regression test for firmware release.
- Designed and developed Watchdog Console Aggregator in C#, a software with web interface and runs as a service on Windows XP/7/Server2008 that aggregates remote sensor data and webcam images from multiple climate monitors in Data Centers and logs data in a SQL database, CE or Express.
- Development tools: Visual Studio, Eclipse, Syslog, WireShark. Tera Term, Emacs, vi, vim, VirtualBox, VMware, CVS, Mercurial, Git, BugZilla, Redmine
- Familiar with HTTP, HTTPS, SSL, SMTP, NTP, DNS, DHCP, ICMP, TELNET, FTP, TFTP, Authentication Certificate, XML, HTML, CSS, MIB.

Process Engineer III, Applied Materials Inc, Sunnyvale, CA, 8/2004 ~ 8/2007

- Developed simulation program (VC++) based on linear regression to predict epi layer profile from the parameters of each heating zone.
- Developed real-time data collection, processing and display program using core Java, Swing.

Graduate Student, Electrical Engineering, University of Texas at Austin, Austin, Texas, 8/1999 ~ 8/2004

- Developed a highly efficient method (non-Monte Carlo) to accurately simulate the gain and excess noise of avalanche process in avalanche photodiodes by solving coupled recursive integral equations. Wrote simulation program in VC++ as standalone Windows application software with graphical user interface. Windows2K, multithread programming, VC++.

EDUCATION

University of Texas at Austin, Ph.D. Electrical & Computer Engineering, 8/1999 ~ 5/2004